

circuited there arises problems. The following are some common protection mechanisms built into power supplies.

Circuit Protection (SCP): If any output from the SMPS is shorted then the SPMS shuts down.

Under Voltage Protection (UVP): There are threshold values assigned to each voltage line and when the value drops below the threshold value, the SMPS shuts down.

Over Voltage Protection (OVP): Similar to under voltage but works in the opposite fashion.

Over Current Protection (OCP): Each voltage line has an upper limit to the current it can supply, when that is exceeded this causes the SMPS to shut down.

Over Power Protection (OPP) or Over Load Protection (OLP): Switches the SMPS off when excess power than rated capacity is drawn.

Over Temperature Protection (OTP): Switches off the SMPS when temperature goes above a certain limit.

Now all the above protection circuits might not be present in all SMPS, but you can figure out just by reading what they do that they are quite important. While SCP, OVP and OCP are quite important, having the rest goes the extra mile. OTP is pretty much common in a lot of modern day circuits.

What to buy

When considering a new SMPS for your PC keep the following points in mind

1. Efficiency matters, higher efficiency power supplies are economical, have better protection mechanisms since they are a bit expensive and are cooler than normal power supplies.
2. Manufacturers rate SMPS a bit conservatively, i.e., a SMPS rated for 850W might be able to provide roughly 1000W but such scenarios should be avoided.
3. Always keep in mind how many connectors you'll need, buying a cheap power supply and daisy chaining everything onto one rail will damage the power supply in the long run.
4. If you are more of a 24x7 computer person then go for a higher wattage SMPS than you need since SMPS capacitors lose their capacities due to overuse so within a year your SMPS's wattage should fall a bit below the rated value.
5. If your SMPS is a pure sine wave SMPS then you should get a compat-